Chatbot Project Documentation (LangChain + Hugging Face)

□ Project Title

AI Chatbot using LangChain and Hugging Face API

□ Introduction

This project is a command-line based AI chatbot built using **LangChain** and **Hugging Face Inference API**. The chatbot interacts with users in real-time, maintains conversation history, and generates intelligent responses using a Large Language Model (LLM).

□ Objective

The main goal of this project is to:

- Build an interactive chatbot using modern AI tools
- Integrate Hugging Face models with LangChain
- Maintain conversational memory
- Evaluate chatbot accuracy using different types of questions

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□ Technologies Used

- Python
- LangChain
- Hugging Face API
- dotenv (for environment variables)
- Streamlit (optional UI support)

□ Working of the Code

1. Environment Setup

- The `.env` file is used to securely store the Hugging Face API token.
- The token is loaded using `load_dotenv()`.

2. Model Initialization

- The chatbot uses the model:
  meta-llama/Llama-3.1-8B-Instruct
- Configurations include:
  - `max_new_tokens = 200` (limits response length)
  - `temperature = 0.2` (low randomness → more accurate responses)

3. Chat Model

- `ChatHuggingFace` is used to wrap the LLM and handle chat-based interactions.

4. Chat History

- A list is used to store:
  - System message (defines assistant behavior)
  - User messages
  - AI responses

5. Conversation Loop

- The chatbot runs in a loop:
  - Takes user input
  - Sends it to the model
  - Stores response in chat history
  - Prints output

6. Exit Condition

- If user types `"exit"`, the chatbot stops execution.

□ Code Explanation (Simplified Flow)

1. Load API key
2. Initialize model
3. Create chat history
4. Take user input
5. Send input to model
6. Receive and display response
7. Repeat until exit

□ Testing & Evaluation

To evaluate chatbot accuracy, different types of questions should be asked:

□ Sample Test Questions

General Knowledge

What is the difference between Artificial Intelligence, Machine Learning, and Deep Learning?

```
You: What is the difference between AI and ML in one line?
Hami:  Artificial Intelligence (AI) is the broader field that enables machines to think and act like humans,
while Machine Learning (ML) is a subset of AI that focuses on training algorithms to learn from data and impr
ove their performance over time.
You: 
```

Logical Reasoning

If a train travels 60 km in 1 hour, how long will it take to travel 150 km at the same speed?

```
You: If a train travels 60km in 1 hour, how long will it take to travel 150km at the same speed?
Hami:  To find the time it takes to travel 150km, we need to divide the total distance by the speed.

Speed = 60km/h
Distance = 150km

Time = Distance / Speed
Time = 150km / 60km/h
Time = 2.5 hours

So, it will take the train 2.5 hours to travel 150km at the same speed.
You: 
```

Data Interpretation

If a company's revenue increases by 20% each year, what will be the total growth after 3 years?

```
You: If a company's revenue increases by 20% each year, what will be the total growth after 3 years?
Hami:  To find the total growth after 3 years, we need to calculate the revenue at the end of each year.

Year 1:
Initial Revenue = 100% (let's assume it's 100 for simplicity)
Revenue after 20% increase = 100 + (100 * 0.20) = 120

Year 2:
Revenue after 20% increase from Year 1 = 120 + (120 * 0.20) = 144

Year 3:
Revenue after 20% increase from Year 2 = 144 + (144 * 0.20) = 172.8

To find the total growth, we can calculate the ratio of the final revenue to the initial revenue:
Total Growth = (Final Revenue / Initial Revenue) - 1
Total Growth = (172.8 / 100) - 1
Total Growth = 0.728
Total Growth = 72.8%

So, the total growth after
You: 
```

Coding Question

Write a Python programm to check whether a number is a prime.

```
You: write a python code to Check the number is prime or not?
Hami:  **Prime Number Checker in Python**
===================================

Here's a simple Python function to check if a number is prime or not:

```python
def is_prime(n):
    """
    Checks if a number is prime or not.

    Args:
        n (int): The number to check.

    Returns:
        bool: True if the number is prime, False otherwise.
    """
    if n <= 1:
        return False
    for i in range(2, int(n ** 0.5) + 1):
        if n % i == 0:
            return False
    return True
```
```

Critical Thinking

Why do you think AI models sometimes generate incorrect or hallucinated responses?

```
You: Why do you think Ai model sometimes generate incorrect or hallucinated responses?
Hami:  AI models, including those based on deep learning and natural language processing (NLP), can sometimes genera
te incorrect or hallucinated responses due to several reasons:

1. **Lack of domain knowledge**: AI models may not have sufficient knowledge or understanding of the domain or topic
 they are being asked about. This can lead to incorrect or irrelevant responses.
2. **Limited training data**: If the training data is limited, biased, or incomplete, the AI model may not learn to
recognize and correct errors or inaccuracies.
3. **Overfitting**: When an AI model is overfitting to the training data, it may become too specialized and fail to
generalize well to new, unseen data, leading to incorrect responses.
4. **Adversarial examples**: AI models can be vulnerable to adversarial examples, which are specifically designed to
 mislead or deceive the model. These examples can cause the model to generate incorrect or hallucinated responses.
```

Real-world Knowledge

What are the main causes of climate change, and how can individuals help reduce it?

```
You: What are the main causes of climate change, and how individuals help reduce it?
Hami:  **Main Causes of Climate Change:**

1. **Greenhouse gases**: The burning of fossil fuels (coal, oil, and gas) releases carbon dioxide (CO2), methane (CH4),
 and other greenhouse gases, which trap heat in the atmosphere and contribute to global warming.
2. **Deforestation**: The clearance of forests for agriculture, urbanization, and other purposes releases carbon stored
 in trees and reduces the ability of forests to act as carbon sinks.
3. **Land use changes**: Changes in land use, such as the conversion of natural habitats to agricultural land or urban
areas, can lead to the release of stored carbon and reduce the ability of ecosystems to sequester carbon.
4. **Agriculture**: The production of meat, especially beef, and other animal products leads to the release of methane
and nitrous oxide (N2O), two potent greenhouse gases.
```

Problem Solving

A man has 3 apples. He gives 1 to his friend and buys 5 more. How many apples does he have now?

```
You: A man has 3 apples. He gives 1 to his friend and buy 5 more. How many apples does have now?
Hami:  To find the total number of apples the man has now, we need to add the number of apples he already had to the
number of apples he bought.

Initial number of apples = 3
Number of apples given to friend = 1
Number of apples bought = 5

Total number of apples = Initial number of apples + Number of apples bought
Total number of apples = 3 + 5
Total number of apples = 8

So, the man now has 8 apples.
```

□ Features

- Maintains conversation context
- Generates human-like responses
- Simple and lightweight implementation
- Easy to extend (UI with Streamlit)

□ Limitations

- Limited token response length
- May generate incorrect (hallucinated) answers
- Depends on API availability
- No advanced memory persistence

□ Future Improvements

- Add Streamlit UI for better user experience
- Implement memory storage (database)
- Add response evaluation metrics
- Improve prompt engineering
- Add multi-language support

□ Conclusion

This project demonstrates how to build a functional AI chatbot using LangChain and Hugging Face. It highlights the integration of LLMs into applications and provides a foundation for building more advanced AI systems.